CS19541 - COMPUTER NETWORKS LAB MANUAL

EXPERIMENT 11

AIM: To configure and verify static routing and RIP routing between networks using Cisco Packet Tracer.

DESCRIPTION:

- **Static Routing:** A routing method where network administrators manually define explicit paths in the router's routing table. Static routing provides precise control over traffic paths, lower processor overhead, and enhanced security, but lacks dynamic scalability. Primary and backup paths (floating static routes) can be configured for fault tolerance.
- **Routing Information Protocol (RIP):** A Distance Vector routing protocol that uses hop count as its metric to determine the best path to a remote destination network. RIP Version 2 supports classless routing (CIDR) and variable-length subnet masks (VLSM).

FEATURES AND CAPABILITIES:

What we can do with Static Routing and RIP:

- Interconnect multiple remote subnets across different routers.
- Define fixed primary routes and backup paths for link failure recovery.

- Enable automatic dynamic path learning across connected routers using RIP advertisements.
- Verify routing tables and trace end-to-end multi-hop communication paths.
- **Use Cases:** Static routing is ideal for small networks or stub networks with fixed gateways. RIP is used in small-to-medium enterprise networks for basic dynamic routing without heavy hardware overhead.

NETWORK TOPOLOGY OVERVIEW:

Devices Used:

- **3 Routers:** Interconnected via Serial or Ethernet interfaces to route packets across multiple networks.
- **PCs:** End-host devices connected to the LAN interface of edge routers.
- **Serial / Ethernet Cables:** Connecting router-to-router (WAN) and router-to-PC (LAN) links.
- **Input Requirements:** Network addresses, subnet masks, next-hop IP addresses, interface IP assignments, and clock rates for DCE serial interfaces.

STEP-BY-STEP PROCEDURE

Task 1: Topology Design & Interface Configuration

1. Drag and drop three routers and required PCs into the Cisco Packet Tracer workspace.
2. Interconnect the routers and PCs using appropriate Straight-Through and Serial DCE/DTE cables.
3. Assign IP addresses and default gateways to all PCs.

4. Configure interface IP addresses on all routers and bring them up using no shutdown.
5. On DCE serial interfaces, set appropriate clock rate values.

Task 2: Configure Static Routing

1. On each router, configure static paths to non-directly connected networks using static routing commands.
2. Configure backup static routes where redundancy is required.
3. Verify static routes using show ip route static.

```
Router1>enable
Router1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router1(config)#interface Serial0/0/0
Router1(config-if)#ip address 192.168.1.1 255.255.255.0
Router1(config-if)#clock rate 64000
Router1(config-if)#no shutdown
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

Router1(config-if)#exit
Router1(config)#ip route 10.1.1.0 255.255.255.0 192.168.1.2
Router1(config)#ip route 10.1.1.0 255.255.255.0 192.168.2.2 10
Router1(config)#end
Router1#show ip route static
Codes: L - local, C - connected, S - static, R - RIP
Gateway of last resort is not set

S      10.1.1.0/24 [1/0] via 192.168.1.2
Router1#
```

Task 3: Configure RIP Routing Protocol

1. Enable RIP routing on each router.
2. Advertise directly connected network addresses.
3. Verify dynamic routes in the routing table using show ip route.

```
Router1>enable
Router1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router1(config)#interface Serial0/0/0
Router1(config-if)#ip address 192.168.1.1 255.255.255.0
Router1(config-if)#clock rate 64000
Router1(config-if)#no shutdown
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

Router1(config-if)#exit
Router1(config)#ip route 10.1.1.0 255.255.255.0 192.168.1.2
Router1(config)#ip route 10.1.1.0 255.255.255.0 192.168.2.2 10
Router1(config)#end
Router1#show ip route static
Codes: L - local, C - connected, S - static, R - RIP
Gateway of last resort is not set

S      10.1.1.0/24 [1/0] via 192.168.1.2
Router1#
```

Task 4: Connectivity & Verification

4. Open the Command Prompt on PC0 and test end-to-end connectivity to PC1.
5. Trace the path taken by packets between PC0 and PC1.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.1.1.10

Pinging 10.1.1.10 with 32 bytes of data:

Reply from 10.1.1.10: bytes=32 time=2ms TTL=126
Reply from 10.1.1.10: bytes=32 time=1ms TTL=126
Reply from 10.1.1.10: bytes=32 time=1ms TTL=126
Reply from 10.1.1.10: bytes=32 time=1ms TTL=126

Ping statistics for 10.1.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\>tracert 10.1.1.10

Tracing route to 10.1.1.10 over a maximum of 30 hops:

  1     1 ms     1 ms     1 ms  192.168.10.1
  2     2 ms     1 ms     1 ms  192.168.1.2
  3     1 ms     1 ms     1 ms  10.1.1.10

Trace complete.

C:\>
```

Result:

Thus the experiment to configure and verify static routing and RIP routing between networks using Cisco Packet Tracer has been studied and executed successfully.